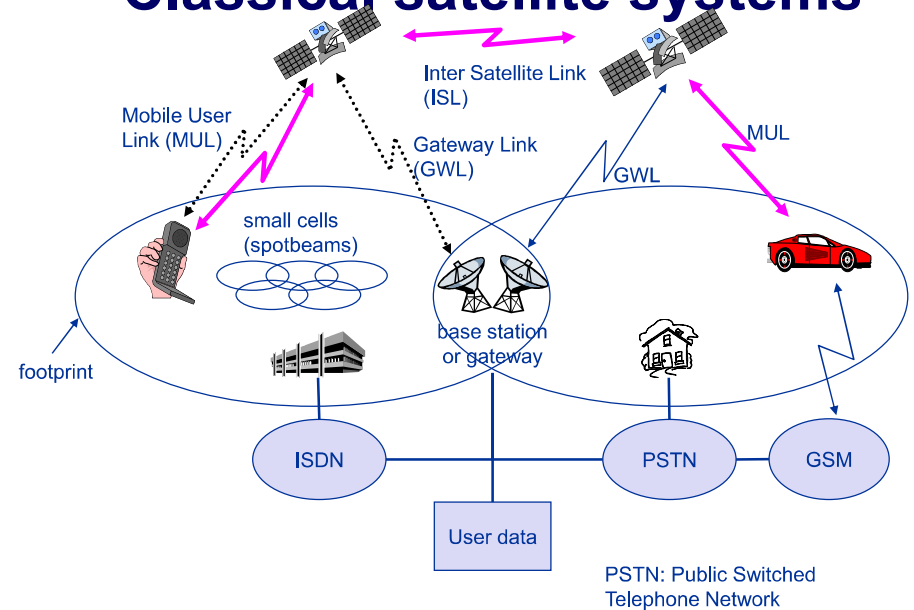


Traditional vs. Modern Satellites

- Traditional satellite were simple transponders
 - Receive a signal on one frequency
 - Amplify it
 - Transmit it on another frequency
- Today: Satellites are like Flying routers
 - Many functions of higher communication layers
 - Inter-satellite routing
 - Error correction
- Earlier only analog amplification
- Now digital regeneration also
 - Satellite decodes the signals into a bit-stream
 - And codes it again into a signal
 - Hence, higher quality of received signal on earth

Classical satellite systems



Basics

- Satellites in circular orbits
 - attractive force of the earth due to gravity

$$F_g = m g (R/r)^2$$
 - Centrifugal force, $F_c = m r \omega^2$
 - m: mass of the satellite
 - R: radius of the earth ($R = 6370 \text{ km}$)
 - r: distance of the satellite to the center of earth
 - g: acceleration of gravity ($g = 9.81 \text{ m/s}^2$)
 - ω : angular velocity, $\omega = 2 \pi f$
 - f: rotation frequency

Basics

- To keep satellite stable in circular orbit
 - $F_g = F_c$
 - $F_g = m g (R/r)^2$
 - $F_c = m r \omega^2$
$$m g (R/r)^2 = m r \omega^2$$
- The distance r of the satellite to the center of the earth results in the following equation:

$$r = \sqrt[3]{\frac{gR^2}{(2\pi f)^2}}$$

Geostationary satellites

- Orbit **35,786 km** distance from earth surface, orbit in equatorial plane (inclination 0°)
- complete rotation **exactly one day**, satellite is synchronous to earth rotation
- fix antenna positions, no adjusting necessary
- A large footprint (up to 34% of earth surface!), therefore difficult to reuse frequencies
- Three satellites can cover the whole globe
- HIGH transmit power needed
- high latency due to long distance (275 ms)
- Not useful for global coverage **for small mobile phones** and data transmission, **typically used for radio and TV transmission**

MEO systems

- Orbit approx. 5000 - 12000 km above earth surface
- Comparison with LEO systems:
 - slower moving satellites
 - less satellites needed
 - simpler system design
 - for many connections no hand-over needed
 - higher latency, about 70 - 80 ms
 - higher sending power needed
 - special antennas for small footprints needed
- Example:
 - GPS: period 12 hrs

LEO systems

- Orbit 500 - 1500 km above earth surface
- visibility of a satellite 95 - 120 minutes
- global radio coverage possible
- latency comparable with terrestrial long distance connections, 5 - 10 ms
- smaller footprints, **better frequency reuse**
- but now **HANDOVER necessary** from one satellite to another
- many satellites necessary for global coverage
- more complex systems due to moving satellites
- Examples:
 - Iridium (start 1998, 66 satellites)
 - Globalstar (start 1999, 48 satellites)

